

Predictive fidelity versus training utility in surrogate-based reinforcement learning for HVAC control: a role-separated hybrid remedy

Abstract

Deep reinforcement learning (RL) controllers for heating, ventilation and air-conditioning (HVAC) systems are increasingly trained against fast neural surrogates of the building, on the implicit assumption that a more accurate surrogate is a better training environment. This paper reports, diagnoses and repairs a counterexample obtained on the BOPTEST single-zone air-based emulator. A calibrated grey-box resistance–capacitance surrogate with a neural residual heat-flow head is by far the better predictor (24 h rollout error 0.644 °C against 1.557 °C for a compact black-box surrogate), yet a proximal-policy-optimization controller trained directly on it collapses on the live emulator for every random seed, spending more than 77% of the time outside the comfort band. Retraining the same black-box surrogate on a finer 15-minute corpus makes it more accurate (0.876 °C) and equally unusable, which identifies the coarse per-step temperature increment, not the model class, as the operative variable. A role-separated hybrid training environment is then proposed: the black-box surrogate supplies the rollout dynamics, while the frozen grey-box twin acts only as a per-step model-disagreement reward censor. The hybrid gives the best worst-window maintenance score of the study (0.087 against 0.095 for the best single-model backend, and 0.041 on the typical window) with comfort violation below 5% on both windows, at about 85 times live-simulator throughput. Boundary tests across three controller families and three hydronic test cases indicate that the censor weight is controller-family specific, and show that the calibration pipeline transfers whereas the frozen policy does not.

Keywords: reinforcement learning, HVAC control, surrogate modeling, digital twin calibration, reward shaping, building energy simulation

1. Introduction

Buildings account for roughly one third of global final energy use and HVAC systems are their single largest consumer [1, 2], which makes supervisory HVAC control a high-leverage decarbonization target. It is also intrinsically difficult: zone thermodynamics are nonlinear and coupled, and the plant is driven by time-varying weather and occupancy. The proportional–integral (PI) and rule-based controllers used in practice are robust but cannot exploit this structure [3, 4], whereas model predictive control can, in principle, but requires an accurate and continuously maintained building model that is expensive to identify and brittle under mismatch [5].

Deep reinforcement learning (DRL) is an attractive alternative because the control law is learned directly from interaction, and standardized runtime environments such as BOPTEST make the resulting controllers reproducibly comparable [6–8]. The practical obstacle is training cost: policy-gradient algorithms consume millions of environment steps, so training on a live building, or on a Modelica simulator stepped through an HTTP interface, is prohibitively slow and risks sustained comfort violations during exploration [9]. The established response is to train against a fast learned surrogate of the building, increasingly a physics-informed digital twin such as a resistance–capacitance (RC) network or a Neural Ordinary Differential Equation model [10–13], which generates transitions orders of magnitude faster than real time [14, 15].

This paradigm rests on an assumption that is rarely made explicit and, to the best of the authors' knowledge, never tested directly in the HVAC setting: that a surrogate with higher predictive fidelity is automatically a better environment in which to train a controller. We find that the assumption can fail, exposing what we call the *fidelity–utility paradox*: the surrogate

with the lowest multi-step rollout error can be the worst environment for policy-gradient search (Fig. 1). Trained directly on the high-fidelity physical twin, the policy converges to a near bang-bang law and fails on the live BOPTEST runtime, even though that twin is the more accurate offline predictor. The driver is measurable on the surrogate side: the higher-fidelity backends expose a much rougher action→next-temperature response surface, which policy-gradient search exploits into a policy that does not survive the distribution shift to the live simulator [16].

Two secondary gaps accompany the central one. HVAC control is partially observable: a small, five-dimensional instantaneous observation is too impoverished for forecast-dependent and preference-conditioned control [4, 17], and a single physics-regularization strength need not suit thermostatic, hierarchical and multi-objective agents alike. Existing studies seldom separate surrogate fidelity from controller utility, rarely scrutinize the observation interface, and almost never pre-specify cross-test-case transfer.

The proposed method is a hybrid, role-separating architecture that decouples prediction from training. A compact control-oriented black-box surrogate (BB) supplies smooth, control-friendly Markov rollout dynamics, while an inverse-calibrated physics-informed twin (GB), whose zone thermal capacitance C_{zon} is identified from telemetry, acts as a *frozen per-step reward-shaping censor*. The censor adds a disagreement penalty to the reward, not a term to the policy loss, and the frozen twin never becomes the rollout model, so the policy keeps learning over the smooth rollout dynamics of BB while state–action regions on which the physical twin disagrees are discouraged. To address partial observability, the controller interface is widened to a 17-dimensional, forecast-augmented direct-supply-temperature observation.

The evidence hierarchy is deliberately narrow. The primary causal test is a four-backend thermostatic-PPO comparison: coarse BB, matched-resolution BB, direct GB, and the hybrid backend. Hierarchical DRL, multi-objective RL and hydronic transfer are not used to enlarge that central claim; they are boundary tests that identify where the role-separated recipe does and does not transfer. Four questions that are often conflated are separated and stated as pre-specified, falsifiable claims, reported whether supported or falsified:

- H1. Fidelity–utility:** a surrogate with higher predictive fidelity is a better RL training environment.
- H2. Role separation:** using the physical twin as a frozen reward-shaping censor over a smooth black-box rollout recovers the control utility that direct use of the twin destroys.
- H3. Censor-weight universality:** a single model-disagreement censor weight transfers across controller families.
- H4. Transferability:** the inverse-calibration pipeline and the frozen controller transfer to related hydronic test cases under documented recalibration.

Stating these in advance lets the negative outcomes be read as findings rather than tuning artifacts. The engineering contributions are threefold. (i) Predictive fidelity is separated from RL training utility: a calibrated twin is the best predictor, but direct training on it collapses in live BOPTEST, and a matched-resolution black-box ablation shows the failure is not a grey-box artifact. (ii) The collapse is diagnosed and repaired by role separation, which recovers the best cross-window robustness of the study at about 85 times live throughput. (iii) The transfer boundary is shown to be component-level: the calibration pipeline transfers across hydronic test cases, whereas frozen-policy transfer is regime-dependent.

Section 2 positions the work; Section 3 defines the control problem, the surrogates, the hybrid and the protocol; Section 4 reports and discusses the results; Section 5 concludes. Reproduction details, hyperparameters, seed tables and additional diagnostics are given in the Supplementary Material.

2. Related work

2.1. Benchmarks and HVAC deep reinforcement learning

Reproducible HVAC-control research depends on a shared benchmark, because controller claims otherwise vary with the simulator, weather year, disturbance schedule and comfort definition. BOPTEST serves Modelica building models through a uniform API [6, 7] with a Gym interface for RL workflows [8], complementing demand-response benchmarks such as CityLearn [18, 19]; every controller here is finally re-scored in BOPTEST, not only in its learned environment. Against this benchmark, HVAC DRL studies compare learned policies with rule-based, Guideline 36 or MPC references [2, 3, 9, 20] and extend them to multi-agent and context-aware settings [21, 22]; hierarchical policies decompose comfort and energy objectives across timescales [23–25], and multi-objective RL represents the energy–comfort trade-off explicitly [17, 26–28]. These works fix the training environment, which is precisely what this study varies.

2.2. Learned building surrogates

Learned building surrogates make DRL-scale training feasible where direct simulation is too slow, spanning black-box networks, grey-box RC models, Neural ODEs and physics-informed learning [10–13, 29–33], and connecting to the broader scientific-machine-learning literature on operator learning and projection-based reduced-order models [34–36]. Surrogate development has itself been framed as a staged validation protocol [14], and fast-surrogate work emphasizes runtime feasibility [15]. Most of this literature validates prediction metrics, yet a controller, like a surrogate-based optimizer that can drive an accurate surrogate into infeasible optima by exploiting its residual error [37], uses the surrogate as a closed-loop transition function under policy-induced states. We therefore report predictive fidelity and runtime first, then test the stronger claim that fidelity implies training utility.

2.3. Objective mismatch and distribution shift

That stronger claim is a distribution-shift question. On-policy data collection and observation design shape the states an RL controller learns on [4, 38, 39], and safe-RL work shows that small model errors become operationally important once actions are optimized against them [9, 40]. This links HVAC control to model exploitation and dataset shift in model-based RL [16, 41], and specifically to the *objective mismatch* between a surrogate trained for predictive likelihood and one that yields a good downstream policy [42, 43]; decision- and value-aware model learning answer this mismatch by retraining the model toward the control objective [44], and the same low-error-yet-poor-transfer effect is central to sim-to-real deployment [45]. Training a policy inside a learned model is the model-based, or Dyna, tradition [46], now scaled to world-model agents [47]. Unlike value-aware retraining, we hold both surrogates fixed and instead *measure* the mismatch mechanism, action-response roughness induced by temporal resolution, and resolve it by *role separation* rather than by changing the surrogate’s training objective.

The shaped reward of Section 3.4 has a recognizable form here, and the relationship is worth stating plainly. Model-ensemble methods keep several dynamics models and use their disagreement to stop the policy exploiting any one model’s errors [48], and offline model-based RL makes the device explicit by subtracting a model-uncertainty term from the reward [49]. Our censor is structurally the same instrument. Two things differ, and they are what this paper contributes. First, disagreement is measured not inside an ensemble of like models fitted to a common objective but between two deliberately unlike ones, a smooth control-oriented black box and a calibrated physics-informed twin, so the penalty carries physical information a homogeneous ensemble lacks. Second, the models are assigned roles rather than averaged: the twin never enters the rollout, so policy search still runs over the smooth dynamics that make it tractable. The mechanism is thus an instance of a known family; what is new is the role assignment and its measured effect on a building-control benchmark.

2.4. Transfer across buildings

Transfer-oriented HVAC work reduces the cost of adapting DRL controllers across buildings, zones and climates [5, 50, 51]. These lines rarely separate two questions kept distinct here:

1. Primary paradox test (the central causal claim)

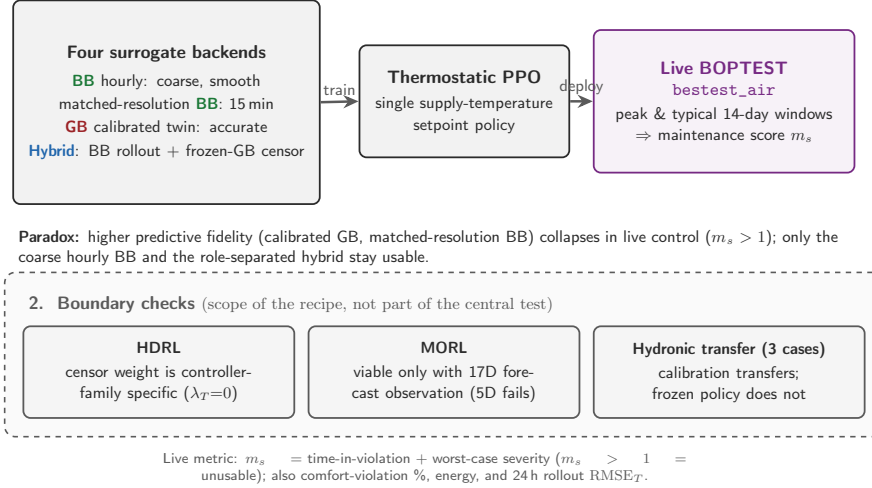


Figure 1: Study overview. *Top, primary paradox test:* four surrogate backends (coarse hourly BB, matched-resolution BB, calibrated GB, and the role-separated hybrid) each train a thermostatic PPO controller scored live on BOPTEST `bestest_air`; the higher-fidelity backends collapse ($m_s > 1$) while the coarse BB and the hybrid stay usable. *Bottom, boundary checks (dashed box):* hierarchical DRL, multi-objective RL and a three-case hydronic transfer study map the scope of the recipe and are *not* part of the central test.

Label	Step	Role in the study
BB hourly	3600 s	Smooth rollout baseline and positive control.
matched BB	900 s	Temporal-resolution ablation isolating whether BB’s utility comes from smoothing rather than black-box structure.
raw GB	900 s	Uncalibrated predictive reference quantifying the calibration gain.
calib. GB	900 s	High-fidelity physical twin; direct-training negative control.
hybrid	900 s	Remedy: BB rollout dynamics plus frozen GB as a model-disagreement reward censor.

Table 1: Experimental taxonomy. The central fidelity–utility test of Table 4 compares BB hourly, matched BB, calibrated GB and the hybrid; raw GB appears only in the predictive comparison of Table 3, to quantify the calibration gain. Hierarchical DRL, multi-objective RL and hydronic transfer are boundary tests.

whether the surrogate recalibrates to the target physics, and whether a frozen controller stays deployable through a new actuator interface, which the version-locked hydronic transfer study of Section 4.6 answers separately.

3. Materials and methods

This section defines the experimental objects: the BOPTEST control problem, the surrogate taxonomy, the hybrid backend, the controller families and the evaluation metrics. Table 1 fixes the naming used throughout. Only the first four backends are part of the central fidelity–utility test.

3.1. Test case and runtime environment

The plant is the BOPTEST `bestest_air` emulator [6, 7], a single-zone office of $6 \times 8 \times 2.7$ m (48 m^2) whose envelope follows the ASHRAE Standard 140 BESTEST Case 900 definition, with heavyweight concrete-block walls, an insulated slab and 12 m^2 of south glazing. Conditioning is an idealized four-pipe

fan-coil unit: a gas boiler of constant efficiency 0.9 feeds the heating coil and a chiller of constant COP 3.0 the cooling coil, with a variable-speed supply fan. The zone is a two-person office occupied 08:00–18:00 with light internal gains and occupied comfort setpoints of 21/24 °C, under Denver-Stapleton TMY weather.

Control is applied through the BOPTEST Run-Time Environment, a Dockerized Modelica emulator with embedded baseline local-loop and supervisory controllers, a disturbance forecaster and a KPI calculator, all exposed through a RESTful HTTP API [8]. The agent does not replace the plant: it *overwrites* one control input, the supply-air temperature setpoint, while the remaining embedded loops run at baseline. At each 900 s control step it reads measurements (zone temperature, CO₂, power) and disturbance forecasts, forms the observation, writes the setpoint through the BOPTEST overwrite pair, and advances the emulator one step; the KPIs are then computed by the runtime from the resulting live trajectory rather than from the surrogate.

The transfer study holds out three single-zone-like hydronic test cases with different heat sources, distribution paths and actuator interfaces (Table 2). They form an increasing-difficulty ladder away from the source case and share the same single-zone first-order energy balance.

3.2. Problem formulation and evaluation metrics

Because the controller never observes the full latent state, control is a partially observable Markov decision process. The agent receives an observation o_t and selects $a_t \sim \pi(\cdot | o_t)$ so as to maximize the expected discounted return [52]

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \right], \quad r_t = -(w_c c_t + w_e e_t), \quad (1)$$

with c_t a thermal-discomfort term, e_t the heating energy over the step, and w_c, w_e fixed trade-off weights. This scalarized comfort–energy objective follows the standard multi-objective treatment of HVAC control, where preferences are represented by explicit weights rather than hidden inside a single black-box

Test case	Type	Heat source / emitter	Climate
bestest_air (source)	Office, 48 m ²	Gas boiler + chiller / 4-pipe fan-coil	Denver
bestest_hydronic	Residential, 48 m ²	Gas water heater / radiator (5 kW)	Brussels
bestest_hydronic_heat_pump	Residential, 192 m ²	Air-to-water heat pump (15 kW) / floor loop	Brussels
singlezone_commercial_hydronic	Commercial, 8500 m ²	District heating / AHU + radiator circuit	Copenhagen

Table 2: BOPTEST test cases used in this study [6]. All are single-zone or single-zone-like and heating-capable. The first is the source case on which the surrogates are built; the other three are held out for the transfer study of Section 4.6.

reward [26–28]. The comfort component is piecewise-linear with an asymmetric, ambient-dependent slope, steeper for cold-weather undershoot and fixed a priori from this physical asymmetry rather than tuned to any backend, plus a small in-band bonus; the energy component is proportional to heating power.

Because HVAC control depends on forecasts, a purely instantaneous five-dimensional observation is insufficient [4]: the canonical interface is a 17-dimensional observation comprising physical states, cyclic time features, ambient forecasts and previous actions. The agent acts through a single normalized supply-temperature command,

$$a_t \in [-1, 1], \quad T_{\text{sup},t} = 18 + \frac{a_t + 1}{2} (35 - 18) \quad [^\circ\text{C}], \quad (2)$$

with a 1.0 °C deadband, a per-step rate limit and a 21–24 °C comfort band. The surrogate transition is *trained* on a two-dimensional command, the supply-temperature action together with a normalized fan setting, both excited only while the offline corpus is generated; during every RL rollout and every live evaluation the fan is held at its BOPTEST baseline and only the supply-temperature setpoint is commanded.

For evaluation, as opposed to training, deployed controllers are scored with the duration-and-severity maintenance score of Wang et al. [9],

$$m_s = r_{\text{time}} + r_{\text{sev}}, \quad (3)$$

where $r_{\text{time}} = N_{\text{viol}}/N \in [0, 1]$ is the fraction of control steps in which the zone temperature leaves the comfort band $[T_{\text{lo}}, T_{\text{hi}}]$ and

$$r_{\text{sev}} = \max_t \max \left((T_{\text{zone},t} - T_{\text{hi}})/T_{\text{hi}}, (T_{\text{lo}} - T_{\text{zone},t})/T_{\text{lo}}, 0 \right)$$

is the largest single-step band exceedance normalized by the band bound. Both terms are dimensionless and lower is better; because $r_{\text{time}} \leq 1$, the threshold $m_s > 1$ flags a controller that is either out of band for a large fraction of the horizon or commits a severe worst-case excursion, the regime treated here as a failed deployment. Separating the shaped training reward (1) from this unshaped safety score follows safe-RL practice [40]. Predictive fidelity is reported as the multi-step rollout RMSE of the zone temperature; control utility as m_s , the comfort-violation fraction and heating energy, always measured on the live emulator and never on the surrogate.

3.3. Surrogate backends

All surrogates approximate the same discrete-time transition $s_{t+1} = \Phi(s_t, a_t, d_t)$ and are exposed behind a single adapter, so that only the internal dynamics change across comparisons, never the control interface or the observation map. Each is fit to BOPTEST-generated transition tuples and used as a *deterministic point-transition* map rather than a learned conditional distribution, which is appropriate where the zone’s conditional dynamics are effectively deterministic given the observed drivers, but does not represent transition uncertainty.

Control-oriented black-box surrogate (BB). BB is a compact black-box neural predictor (8482 parameters) of a temperature increment and a power head, $\hat{T}_{t+1} = T_t + f_{\theta,T}(x_t)$ and $\hat{P}_{t+1} = f_{\theta,P}(x_t)$ with $\hat{P} \geq 0$, trained by a multi-horizon (1-, 2-, 4-step) supervised rollout objective with a physical-consistency penalty. Its design goal is not long-horizon accuracy but a *smooth, well-conditioned* response surface for policy-gradient search. Its input vector is deliberately low-dimensional: thermal state, outdoor disturbance, cyclic time embedding and the direct supply-temperature action [20]. The canonical checkpoint is the early-stopped optimum of a 215-epoch run with held-out one-step $R^2 = 0.991$. Its transition carries *no* explicit time step, so the increment learned on the canonical 3600 s corpus is applied unchanged at the 900 s runtime step.

Physics-informed grey-box surrogate (GB). GB replaces the black box with a first-principles lumped-capacity zone balance whose unmodeled heat flow is supplied by a neural residual head,

$$C_{\text{zon}} \frac{dT_{\text{zone}}}{dt} = \dot{Q}(x; \phi), \quad \dot{Q}(x; \phi) = q_{\text{scale}} q_{\phi}(x), \quad (4)$$

with $q_{\text{scale}} = 3000 \text{ W}$, integrated by a clipped explicit Euler step at the control period, and with C_{zon} positively reparameterized and bounded below by $5 \times 10^4 \text{ J/K}$. Embedding this conservation law makes the model physics-informed in the sense of [12, 13, 32, 33], so C_{zon} is physically interpretable, although the analysis treats it as an *effective* capacitance that may absorb unmodeled fast dynamics rather than a direct measurement of zone air capacitance. Because C_{zon} is a property of the building rather than of the data-generating policy, it is expected to remain meaningful across test cases, which is what makes the inverse identification below worthwhile.

The canonical GB is calibrated by a staged inverse identification [10]: Stage A aligns telemetry and removes latency and bias; Stage B identifies C_{zon} as a maximum-a-posteriori estimate under a weak Gaussian prior, restricted to excitation-rich transients where the sensitivity $\partial \hat{T} / \partial C$ is large, since quasi-steady windows would merely pull the estimate back to the prior; and Stage C refines the neural residual heads with C_{zon} frozen. The identified $C_{\text{zon}} = 4.413 \times 10^5 \text{ J/K}$ is a +5.1% update over the physical prior, within one standard error of the Laplace/Fisher interval, and corresponds to about 439 kg of air, the expected order for this zone; a supply-temperature sweep at fixed state gives the physically correct sign in 100% of 400 sampled states.

A third backend, matched-resolution BB, is the *same* black-box architecture retrained on a 900 s corpus. It exists solely to isolate the model-class confound in Section 4.2.

3.4. Role-separated hybrid training environment

The hybrid backend is not a new RL algorithm. It is an application-specific role-separation strategy grounded in three

existing lines of work: model-based and world-model training in learned environments [46, 47], physics-informed and grey-box surrogate modeling [10, 12, 33], and safe-RL reward shaping [9, 40]. The contribution is the allocation of roles inside surrogate-trained HVAC control (Fig. 2): BB supplies the smooth rollout dynamics, while a *frozen* GB acts only as a per-step reward-shaping censor. Training uses the shaped reward

$$\tilde{r}_t = r_t - \lambda_T \delta_{T,t} - \lambda_P \delta_{P,t}, \quad (5)$$

where the disagreement terms are the per-step L_1 distances between the one-step predictions of the rollout surrogate f_θ (BB) and the frozen twin g_ϕ (GB), separately for the zone-temperature and HVAC power channels,

$$\delta_{\bullet,t} = |f_{\theta,\bullet}(s_t, a_t, d_t) - g_{\phi,\bullet}(s_t, a_t, d_t)|,$$

with no squaring, normalization or clipping. The two censor weights differ in scale because they multiply a temperature residual in $^\circ\text{C}$ and a power residual in W ; for the thermostatic controller $\lambda_T = 0.10$ and $\lambda_P = 5 \times 10^{-5}$.

Crucially, g_ϕ is evaluated *forward only* and its outputs are detached, so the censor enters the *reward* alone, never the policy loss, and never the rollout dynamics, which remain those of BB. Because a policy-gradient method estimates its gradient from sampled returns rather than by differentiating the environment, this role split does not preserve any environment-to-policy gradient; what it preserves is the smooth trajectory-and-return distribution of BB, and its favorable optimization conditioning, while penalizing state-action regions on which the frozen twin disagrees with the rollout. In other words, GB does not replace BB, form an accuracy-oriented ensemble, or introduce a second policy-loss term. The penalty measures disagreement between two imperfect surrogates, not violation of a conservation law or of a physically feasible set; it is therefore a *model-disagreement* censor, physics-informed through GB’s structure but not a hard physical-feasibility filter.

3.5. Controller families and protocol

All policies are optimized with proximal policy optimization (PPO) [53] using generalized advantage estimation [54] and the Stable-Baselines3 implementation [55]. Shared settings are learning rate 3×10^{-4} , discount factor $\gamma = 0.99$ (an effective horizon of about 25 h at the 900 s step) and ten optimization epochs per rollout; the families differ in rollout length, minibatch size and total timestep budget. The recipe is evaluated across three controller families of increasing structure: a *thermostatic* PPO agent on the shared forecast-augmented observation; a *hierarchical* DRL controller (HDRL) that decomposes control across timescales in the options/feudal tradition [24, 25] as recently applied to year-round HVAC operation [23]; and a *preference-conditioned* multi-objective controller (MORL) [17, 26].

Policies are trained on their surrogate backend and then evaluated *zero-shot* in closed loop on the live emulator over two targeted 14-day windows: a January peak-heat window with daily-mean ambient -24.4°C and a February typical-heat window at 2.4°C . Because a policy trained on a surrogate is deployed on a different, live system, the central methodological risk is distribution shift [41]; the evaluation therefore always reports the live-emulator score. Each hypothesis, pass threshold and recalibration adapter was pre-specified *before* the corresponding runs.

Implementation. Surrogates use three corpora generated by scripted excitation: a canonical BB corpus (51 200 rows at 1 h), a 15-minute exploration corpus (48 384 rows at 900 s) and a GB calibration corpus (10 744 rows at 900 s); the deliberate one-hour/fifteen-minute pairing is the instrument behind the matched-corpus ablation. All controllers are trained in Stable-Baselines3 2.1.0 under Python 3.11.9. Experiments ran on a laptop workstation (6-core CPU, 32 GB RAM, laptop-class GPU) with the emulator in a Docker container. Behind one HTTP contract sit four interchangeable backends (live BOPTEST, BB, GB, hybrid), so only the dynamics backend changes across the surrogate-versus-live comparison; the surrogate backends run 85.0–220.2 times faster than the live loop, so a 5×10^6 -step policy-optimization run costs about 47 minutes of environment time against roughly 66 hours on the live runtime. Canonical KPI tables use seed 42; the four central backends additionally carry an $N = 3$ seed-robustness audit over seeds {42, 43, 44}, MORL is evaluated over five fixed seeds, and the HDRL censor-weight sweep is replicated over the same three seeds at each of its four λ_T settings.

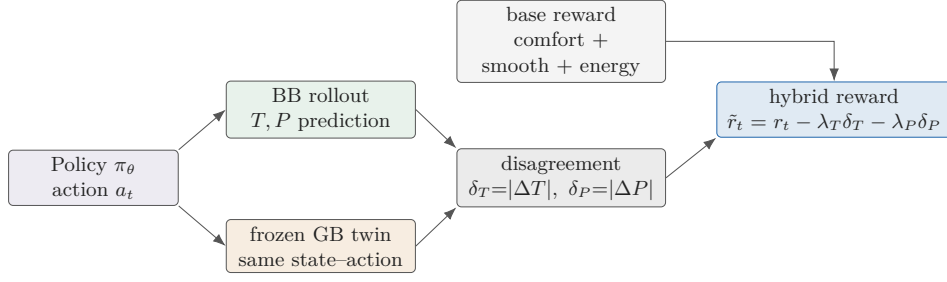
4. Results and discussion

The experiments answer four questions in sequence. *First*, which surrogate is the more faithful predictor? The calibrated GB, by a wide margin (Section 4.1). *Second*, does that higher fidelity make it the better RL training environment? No: used directly, both GB and a matched-resolution BB collapse on the live runtime (Section 4.2), which is the paper’s central causal test. *Third*, why do the high-fidelity backends collapse, and how is control recovered (Sections 4.3–4.4)? *Fourth*, how far does the result transfer (Sections 4.5–4.6)?

4.1. Predictive fidelity: the calibrated twin is the better model

Predictive accuracy is reported as the multi-step rollout $\text{RMSE}_T(h)$ of the per-step error $\hat{T}_i(t+h) - T_i(t+h)$, together with the engineering-tolerance error CDF and its 95th-percentile tail; the coefficient of determination is taken against the held-out mean, so a *negative* 24 h R^2 means the rollout is worse than predicting that mean. Over 24 h rollouts the calibrated GB reaches $\text{RMSE}_T = 0.644^\circ\text{C}$, against 1.466°C for the uncalibrated GB and 1.557°C for the canonical hourly BB (Fig. 3). The hourly BB has a *negative* 24 h R^2 , worse than the historical mean over long horizons and worse than a naive persistence forecast at 24 h (1.557 against 0.969°C), yet it remains a usable control surrogate because policy-gradient search needs smooth local gradients, not long-horizon realism. Calibrated GB instead keeps a positive 24 h R^2 , bounds its tail (24 h P95 1.207°C) and beats persistence at every horizon. The episode-level diagnostic matters because a low aggregate error can hide a single unstable rollout: the calibrated GB stays below the 1°C engineering reference in all eight held-out episodes, with a best episode of 0.486°C and a worst of 0.964°C , which supports predictive validity as a repeated-episode result rather than a single-window artifact.

Because the original BB and the canonical GB artifacts use different data resolutions, a corpus-matched retraining step is required: without it, a reviewer could correctly attribute the GB gain to the 15-minute corpus rather than to Stage A/B/C calibration. Retraining the same black-box architecture on the matched



The frozen GB is a forward-only reward-shaping censor: it is not a policy-loss term and not the rollout model.

Figure 2: Hybrid reward-shaping mechanism. The black-box surrogate (BB) supplies the smooth rollout dynamics; the frozen grey-box surrogate (GB) is evaluated forward-only on the same state-action pair and enters *only* as a per-step reward penalty on BB-GB disagreement, Eq. (5). GB is neither the rollout model nor a policy-loss term.

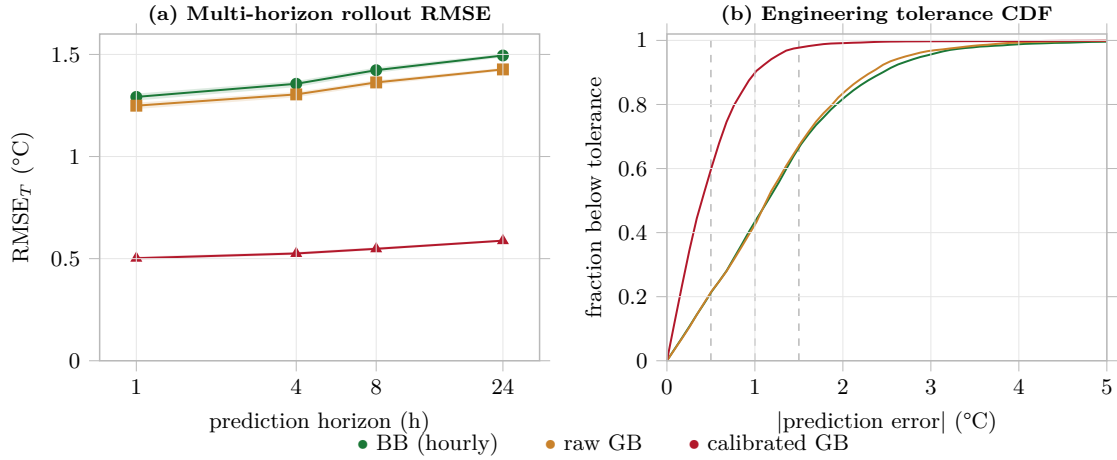


Figure 3: Multi-horizon predictive validity, residual spread and engineering error CDFs from BOPTEST rollouts. The calibrated grey-box twin is the better predictor at every horizon; the coarse black-box surrogate degrades beyond a few hours yet remains the usable training environment.

Variant	Step (s)	Calibration	24 h RMSE _T
BB hourly	3600	n/a (black box)	1.557
BB 15-min matched	900	n/a (black box)	0.876
raw GB	900	OFF	1.466
calibrated GB	900	FULL (A + B + C)	0.644

Table 3: Corpus-matched comparison. The second row is what separates the corpus-resolution contribution from the Stage A/B/C calibration contribution.

corpus brings it to 0.876 °C (Table 3). Of the total 0.913 °C improvement from hourly BB to calibrated GB, 0.681 °C (74.6%) is the corpus shift and 0.232 °C (25.4%) the staged calibration; an alternative split through the uncalibrated GB backbone gives 9.9%/90.1%, the two differing because corpus and architecture interact non-additively. The calibration claim is therefore positive but bounded, and it is reported symmetrically rather than as a single number.

The runtime benchmark closes the predictive-validation step: the calibrated GB is 114.2× faster than the live HTTP loop and the hybrid reference 85.0×, so the surrogate family is not only more accurate after calibration but fast enough to support repeated policy-training experiments.

4.2. Higher predictive fidelity does not imply training utility

Used directly as PPO rollout environments, both the more accurate GB and a matched-resolution BB collapse on the live

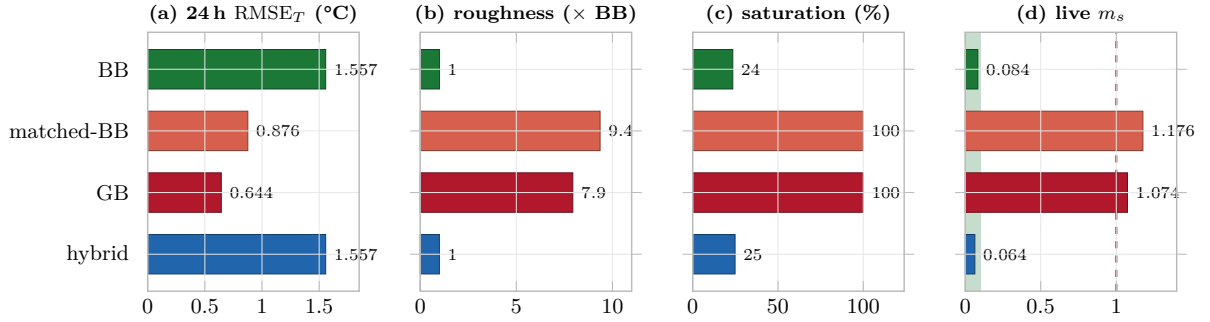
runtime, whereas the less accurate hourly BB trains a usable controller. This is the central causal test, and it compares four training backends under the same thermostatic PPO controller. Direct GB is a *negative control*, not a failed implementation: it asks whether the highest-fidelity predictive surrogate can be used directly as the policy rollout environment.

Table 4 sets predictive fidelity against live control utility, Fig. 4 shows the measured evidence chain and Fig. 5 plots fidelity against utility directly. The result contradicts the working assumption of the surrogate-training paradigm. Pure BB PPO is already usable ($m_s = 0.073$ peak, 0.095 typical). Direct GB PPO fails catastrophically despite its superior predictive fidelity: live violation reaches 77.1% on the peak window and 82.4% on the typical window, with closed-loop RMSE above 4.3 °C. This is not a grey-box artifact: the matched-resolution BB, that is the *same* black-box architecture made strictly more accurate, collapses as well ($m_s = 1.142/1.211$, violation 85.6/91.4%). The hybrid backend resolves the conflict, reaching $m_s = 0.087$ on the peak window and 0.041 on the typical window with violation below 5% on both and lower energy than pure BB on the peak window.

The decomposition $m_s = r_{\text{time}} + r_{\text{sev}}$ exposes the failure structure directly (Fig. 6). Direct GB’s peak score is dominated by $r_{\text{time}} \approx 0.77$, that is out of band 77% of the time, plus a large worst-case severity $r_{\text{sev}} \approx 0.28$; the hybrid’s peak score of 0.087 is almost entirely a small severity term with $r_{\text{time}} < 0.05$. The

Backend	Step (s)	24 h RMSE	m_s (canonical seed $\pm$ seed s.d.)		Violation (%)	
			peak	typical	peak	typical
BB hourly	3600	1.557	0.073 ± 0.019	0.095 ± 0.047	1.5	4.4
matched BB	900	0.876	1.142 ± 0.000	1.211 ± 0.000	85.6	91.4
calibrated GB	900	0.644	1.046 ± 0.022	1.102 ± 0.026	77.1	82.4
hybrid	900	—	0.087 ± 0.026	0.041 ± 0.024	4.7	2.4

Table 4: Central fidelity–utility paradox. Each backend trains the same thermostatic PPO controller, evaluated zero-shot in closed loop on the live BOPTEST runtime over the peak/typical 14-day windows. Predictive fidelity (24 h rollout RMSE, °C) is set against live control utility (m_s , unusable when $m_s > 1$, and comfort violation). Point values are the canonical seed-42 runs, dispersions are standard deviations over seeds {42,43,44}; the usable-versus-collapse verdict spans more than an order of magnitude and is not a seed effect. Averaged over the three seeds the hybrid leads pure BB on *both* windows (0.060 against 0.072 peak, 0.014 against 0.046 typical), so the canonical-seed peak ordering here is a single-seed artifact. The hybrid rolls out on BB and has no independent predictive RMSE.



Quantified evidence chain: lower predictive RMSE_T does not imply lower live m_s

Figure 4: The fidelity–utility paradox at a glance. For the four thermostatic-PPO backends the measured chain runs (A) 24 h rollout RMSE → (B) action-surface roughness → (C) policy saturation → (D) live maintenance score. The higher-fidelity backends win on (A) yet are rougher, saturate and collapse in (D), and the contradiction between (A) and (D) is the paradox. For the hybrid, panel (A) shows the BB rollout RMSE because the hybrid rolls out on BB; panel (D) reports the cross-window mean.

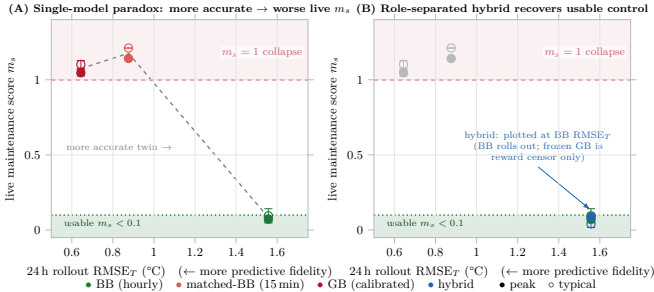


Figure 5: Predictive fidelity against live control utility: a lower rollout RMSE can raise the live maintenance score, and the role-separated hybrid returns to the usable region. Filled and open markers denote the peak and typical windows; error bars are ± 1 seed standard deviation over three seeds.

hybrid therefore does not merely lower the average error, it removes the sustained band departures that dominate the direct-GB score.

Retrained on three fixed seeds, each backend keeps its regime: pure BB and the hybrid hold every seed below the 5% violation bar on both windows, while the matched-resolution BB collapses on every seed ($m_s = 1.14/1.21$, standard deviation below 0.001) and direct GB collapses on every seed ($m_s = 1.07 \pm 0.02 / 1.13 \pm 0.03$). The usable-versus-collapse verdicts are therefore seed-stable, and the single-seed peak-window edge of pure BB over the hybrid does not persist across seeds.

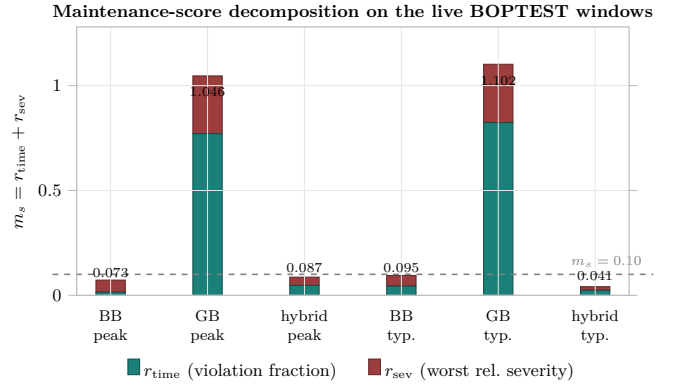


Figure 6: Live BOPTEST maintenance-score decomposition. Direct GB fails through sustained out-of-band time; pure BB and the hybrid keep both duration and severity small.

4.3. Mechanism: temporal coarseness, roughness and saturation

The high-fidelity backends collapse because their action-response surfaces are rougher, which drives near bang-bang policy saturation and live failure. The mechanism is a measured chain from surrogate response geometry to policy behavior to live outcome (Fig. 7). Sweeping each backend’s one-step map $\hat{T}(a_0)$ over a grid of states and computing a scale-free *relative roughness*, mean curvature divided by mean slope, orders the backends cleanly with usability: the usable hourly BB is smoothest (0.17), while the matched-resolution BB and GB are $9.4\times$ and $7.9\times$ rougher. Both collapsing backends saturate the actuator, placing extreme actions in temperature-error regimes

the live building does not support, in 100% of steps against 24% for the usable controller. The time-domain signature is direct GB saturating and leaving the comfort band while the coarse BB and the hybrid stay in band (Fig. 8).

Relative roughness is necessary but not sufficient. A constant Δt rescaling preserves the curvature-to-slope ratio, so a Δt -rescaled hourly BB has the *same* relative roughness of 0.17 as the usable hourly BB, yet it also collapses on every seed ($m_s = 1.19/1.22$). The operative surrogate property is therefore the coarse per-step increment magnitude, of which relative roughness measures one facet, rather than scale-free smoothness alone. This also settles which part of the coarse-versus-fine contrast is load-bearing: the hourly BB is deployed with its hourly increment applied unchanged at the 900 s step, four times the dimensionally consistent per-step change, so part of both its smoothness and its 1.557 °C error is a train/deploy timestep mismatch rather than corpus resolution alone.

Consequently the outcome across the three single-model backends is not a monotonic fidelity–utility curve but a two-level contrast: the temporally coarse backend is usable and both finer backends collapse, with a non-monotonic ordering inside the collapse region. The engineering reading is that policy-gradient search needs an optimization-friendly training landscape as well as accurate predictions, and that the two requirements are traded against one another by the control-step resolution of the training corpus. This is interpreted as an instance of model exploitation in model-based RL, a distribution-shift effect [16, 41]: the sharper response surface of the high-fidelity backends is exploited by policy-gradient search into a near bang-bang law that does not survive transfer to the live plant.

4.4. Role separation restores control

Separating the two roles resolves the conflict. With BB retained as the rollout model and the frozen GB entering only through the censor of Eq. (5), the hybrid attains $m_s = 0.087$ and 0.041 on the peak and typical windows, with comfort violation below 5% on both and lower energy than pure BB on the peak window, at 85.0× live throughput. On the canonical seed pure BB is marginally ahead on the peak window alone; the hybrid has the better *worst* window, and the ordering reverses on both windows once averaged over seeds (Table 4).

The canonical operating point was selected by a thermostatic sweep over $\lambda_T \in \{0.05, 0.10, 0.15\}$ at fixed $\lambda_P = 5 \times 10^{-5}$; the mid setting retains the energy advantage while avoiding the stronger comfort degradation seen at the weaker and stronger censor settings. Only the canonical point’s live KPIs are retained as a frozen artifact: the bracketing settings served selection only and were never carried to the full two-window live protocol, so the headline results of Table 4 are not reported on data used to choose λ_T . The power weight λ_P was not swept: it was fixed a priori at the value that puts the power residual in W on the scale of the temperature residual in °C. Across the hybrid traces the mean temperature disagreement is 0.92 °C on the peak window and 1.01 °C on the typical window (95th percentiles 2.22 and 2.52 °C), and the mean power disagreement 682 W and 735 W respectively: large enough to shape learning and bounded enough to stay informative.

The hybrid should not be read as an accuracy-oriented ensemble of two predictors. If it were, GB would naturally dominate the rollout; the experiments show the opposite, since the physically richer model becomes useful only once its role is restricted

to a detached model-disagreement censor, while the smoother BB remains the training substrate. A second negative control confirms this. Using GB as a policy *initializer* instead, by pre-training on GB and warm-start fine-tuning on the live runtime, is compared against an otherwise identical scratch fine-tune: it does not help, and warm-starting raises m_s by a factor of two to three on both windows. The physical twin belongs in the reward, not in the rollout dynamics and not in the weights.

Transfer-gap diagnostics confirm the same ordering from the surrogate side. Direct GB shows by far the largest surrogate-to-live mismatch, a learned bang-bang exploit whose policy sits at the opposite action bound from BOPTEST throughout, with action-gap norm $g_a \approx 2.0$, while the hybrid shows the smallest gap and stays consistent with the live runtime longest.

4.5. Boundary test I: controller families

The preceding sections establish the paradox and its remedy within the thermostatic PPO family. Two further families test whether the recipe transfers; they are boundary tests, not additional proof of the paradox.

Hierarchical DRL. A seasonal hierarchy routes control to winter and summer PPO setpoint specialists that share the 17-dimensional observation and the supply-temperature action. The thermostatic-optimal censor weight $\lambda_T = 0.10$ does *not* transfer: because each specialist is already comfort-aware, the GB temperature censor over-regularizes the low-level loop and m_s degrades monotonically with λ_T , with peak violation roughly quadrupling from $6.3 \pm 1.7\%$ to $24.4 \pm 3.8\%$ (mean $\pm$ s.d. over seeds {42, 43, 44}). HDRL is best at $\lambda_T = 0$, keeping only the power channel. The correct censor strength is therefore family-specific, and a single “best” regularization strength cannot be reported without naming its controller family. The sweep is seed-replicated at each of its four settings: m_s rises from 0.182 ± 0.017 to 0.442 ± 0.033 on the peak window and from 0.245 ± 0.017 to 0.570 ± 0.077 on the typical window, an endpoint separation of 10.0 and 5.8 pooled seed standard deviations with disjoint ± 1 s.d. bands. The ordering is monotone in the seed means at every step, but the response saturates: the $0.05 \rightarrow 0.10$ increment is only 1.4 pooled standard deviations on both windows and is not resolved above seed noise, so the supported statement is that the censor degrades hierarchical control across the swept range, not that every increment is separately resolvable. The evidence against censor-weight universality is therefore two independently swept families, thermostatic PPO and HDRL, and on the HDRL side it is now seed-replicated; H3 is accordingly reported as *falsified*.

Multi-objective RL. A four-stage MORL pipeline (surrogate pretraining, preference adaptation, live fine-tuning, yearly evaluation) on the power-only hybrid backend is viable only once the observation is widened from five to 17 dimensions: the five-dimensional interface fails ($\text{RMSE}_T = 2.721$ °C, 37.8% violation, $m_s = 0.680$) while the forecast-augmented 17-dimensional vector recovers a usable policy ($\text{RMSE}_T = 0.720$ °C, 4.9%, $m_s = 0.099$). The bottleneck is observation geometry, not the reward scalarization, consistent with forecast-augmented RL HVAC control [4]. MORL yields a usable comfort–energy Pareto front, but an $N = 5$ seed extension turns the canonical operating points into wide distributions (coefficient of variation 0.42–0.61) that fail a pre-specified stability test: promising in the mean, but not deployment-stable without explicit stabilization.

Measured roughness mechanism: rougher action surface $\rightarrow$ policy saturation $\rightarrow$ live collapse (association, not a proven causal law)

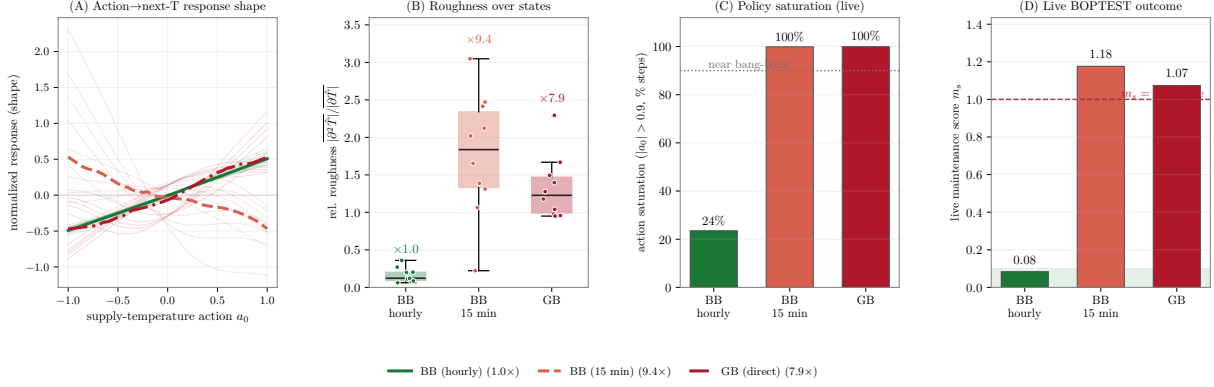


Figure 7: Measured roughness mechanism. The collapsing single-model backends expose rougher action–response surfaces, produce near bang-bang saturation and fail on the live emulator; the pattern is mechanistically consistent with the fidelity–utility paradox but does not by itself prove full loss-landscape causality.

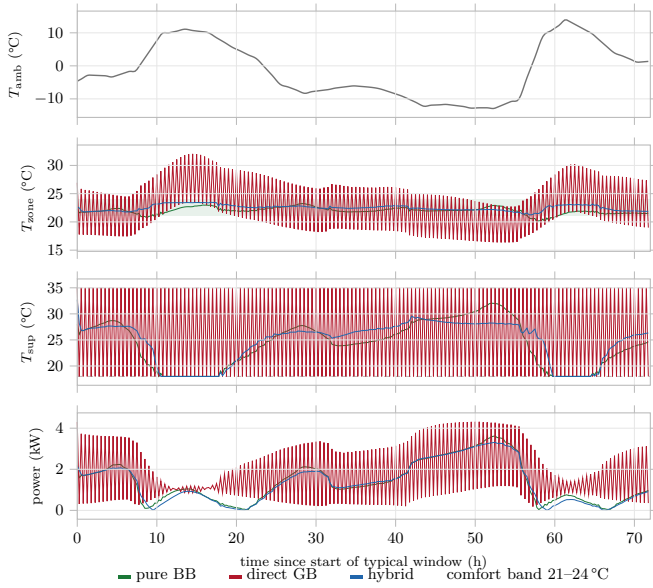


Figure 8: Live closed-loop behavior on the typical window: zone temperature against the comfort band, with ambient disturbance and HVAC power, for the pure-BB, direct-GB and hybrid controllers. Direct GB saturates the actuator and leaves the band; the coarse BB and the hybrid track it.

Synthesis against the PI reference. The BOPTEST built-in PI controller is a reproducible reference, comfort-poor but energy-frugal, with 12-month means $RMSE_T = 3.40^\circ\text{C}$, 63.6% violation and $m_s = 0.910$. All evaluated learned agents reach m_s well below PI, dominating it on comfort at comparable or lower energy. The cross-family maintenance-score decomposition makes the pattern explicit: the failures, direct GB and PI, carry a large time-in-violation term $r_{time} \gtrsim 0.6$, PI by chronic under-heating and direct GB by saturated overshoot, whereas the usable controllers keep both terms small.

4.6. Boundary test II: cross-building transfer

Transfer is decomposed into two components, frozen-policy transfer and surrogate-recalibration transfer, which prevents a misleading binary conclusion. The claim is deliberately bounded: three pre-selected single-zone hydronic cases, documented actuator adapters, pre-specified recalibration, and target-test-case controller fine-tuning explicitly excluded. A pre-specified per-test-case adapter maps the frozen policy’s supply-temperature

command to each target’s BOPTEST override inputs, committed and smoke-tested for directional validity before any run. The three targets differ in heat source, distribution path and actuator set but share the same single-zone first-order energy balance, so transfer changes the *delivery path* and the actuator interface while C_{zon} remains the shared lumped coordinate that Stage B re-identifies. The controller-side pass threshold is normalized by each test case’s built-in PI baseline, $\tau = 1.25 m_{s,PI}$, with the factor pre-specified rather than chosen post hoc; energy is reported on a separate axis, because the commercial case passes the m_s bound while consuming substantially more energy than PI.

Table 5 contains two stories. The surrogate component transfers strongly: full Stage A/B/C recalibration improves rollout RMSE by 60.2–87.8% on all three targets, and the re-identified capacitance clusters at $\rho_C = 1.918 \pm 0.032$ times the source value across an order-of-magnitude change in floor area, read as an *effective* single-zone RC signature rather than a claimed physical invariant. The frozen controller does not transfer uniformly: both residential cases fail the $1.25 \times$ -PI threshold while *saving* energy, and the commercial stretch case passes the threshold at a 35.3% energy penalty. Mean delivered power exposes the regime: about 0.7–2.8 kW for the residential cases, conduction and slow-hydronic dominated, but 76.9 kW for the commercial case, ventilation and AHU dominated, about $28\times$ the heat-pump target, which is the physical reason the commercial controller passes comfort yet overshoots energy. No frozen-policy transfer is deployment-ready (Fig. 9). This component-level boundary is consistent with building heterogeneity requiring local adaptation rather than zero-shot reuse [50, 51]. For practice, the transferable asset is the calibration pipeline, not the frozen control law.

4.7. Limitations

Evaluation is simulation-only on a single-zone emulator, with multi-zone and richer topologies untested and transfer limited to a hydronic family. Several caveats bound the central claim. First, the fidelity–utility relationship is established at three discrete operating points, hourly BB, matched 15-min BB and calibrated GB, rather than as a continuous curve; the three-seed replication shows all verdicts are seed-stable, but an intermediate-resolution sweep is left to future work. Second, the only comparison baseline is BOPTEST’s built-in PI, a relatively weak reference, so controller utility is established against a mod-

Test case	$m_{s,RL}$	$m_{s,PI}$	τ	Ctrl.	$\Delta E\%$	Raw RMSE	Full RMSE	$G\%$	ρ_C
hydronic heat pump	0.665	0.464	0.579	FAIL	-7.3	1.421	0.565	60.2	1.892
hydronic (boiler)	0.976	0.750	0.938	FAIL	-5.8	2.666	0.335	87.4	1.954
commercial hydronic	0.431	0.628	0.785	PASS [†]	35.3	1.952	0.238	87.8	1.909

Table 5: Transferability matrix. $m_{s,RL}$ and $m_{s,PI}$ are the maintenance scores of the frozen RL controller and the built-in PI baseline; the pass threshold is $\tau = 1.25 m_{s,PI}$ and the controller verdict is PASS when $m_{s,RL} \leq \tau$; $\Delta E = 100(E_{RL} - E_{PI})/E_{PI}$; raw and full RMSE ($^{\circ}C$) are 24 h rollout errors before and after staged recalibration, with relative gain G ; ρ_C is the re-identified capacitance ratio against the source value 4.413×10^5 J/K. [†] threshold pass with an energy penalty.

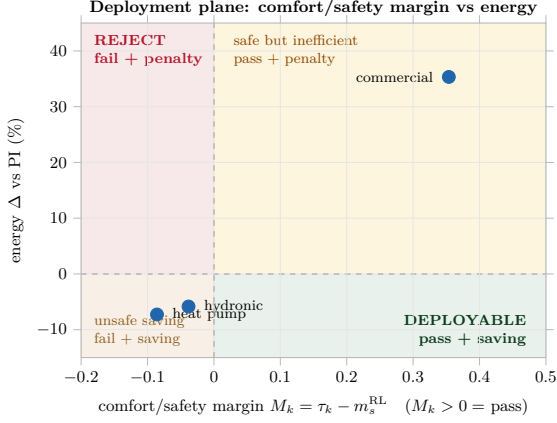


Figure 9: Comfort-energy deployment plane. Residential hydronic cases save energy but fail the comfort threshold; the commercial case passes the threshold with a 35.3% energy penalty. No frozen-policy transfer is deployment-ready.

est rather than a strong tuned rule-based, Guideline-36 or MPC controller on the same RC model. Third, the surrogate power channel is reproduced less accurately than temperature, with a calibrated power-head mean absolute error of 482 W, about 9% of $P_{\max} = 5500$ W; the reported energy KPIs and the transfer ΔE are live-versus-live measurements that do not inherit this error, but the weaker power channel limits predictive-fidelity claims about the power head and adds noise to the training-time energy reward. Fourth, the evidence against H3 now rests on a seed-replicated four-point HDRL sweep ($N = 3$ per setting), but the sweep density remains non-uniform across families: two measured endpoints for thermostatic PPO and a single adopted configuration for MORL. What it establishes is that the thermostatic optimum does not transfer to the hierarchical family, not a fully swept per-family optimum for every family; and within the HDRL sweep the response saturates above $\lambda_T = 0.05$, so the ranking of the two largest weights is not resolved above seed noise. Finally, the mechanism is *measured* statically on each surrogate’s one-step action map and corroborated by the matched-resolution ablation, but a loss-landscape analysis along the training trajectory and generality to off-policy algorithms remain open.

5. Conclusions

This study asked whether a surrogate with higher predictive fidelity is automatically a better environment in which to train a reinforcement-learning HVAC controller, and tested the question across three controller families and a hydronic family of transfer test cases under a pre-specified protocol. The answer is no. Only the temporally coarse, hourly-trained black-box surrogate trains a usable controller: the most accurate physical twin ($0.644^{\circ}C$ 24-hour rollout RMSE) collapses on every seed

with comfort violation above 77%, and the *same* black-box architecture retrained at the finer 900 s resolution becomes a *more* accurate predictor ($0.876^{\circ}C$) yet also fails as a training environment ($m_s = 1.14/1.21$, violation above 85%). H1 is therefore falsified, and the operative variable is the surrogate’s coarse per-step increment magnitude, that is the temporal resolution of the training corpus, not the surrogate model class.

Separating the two roles resolves it. Using the black-box surrogate for smooth rollout dynamics and a frozen physical twin as a reward-shaping censor (H2 supported) gives the best worst-window maintenance score of the study ($m_s = 0.087$ peak and 0.041 typical, against $0.073/0.095$ for pure BB), with violation below 5% on both windows, at an $85\times$ training speed-up. The censor strength is controller-family specific ($\lambda_T = 0.10$ for thermostatic PPO against 0 for the hierarchical agent), so H3 is falsified on seed-replicated evidence ($N = 3$ at each of four swept weights), on the same footing as the verdicts on H1 and H2. Transferability resolves into a component-level boundary (H4): the inverse-calibration pipeline transfers, with rollout RMSE falling by 60.2–87.8% and the zone capacitance re-identifying at a near-uniform ratio, whereas the frozen control policy does not transfer zero-shot.

The contribution is therefore less a new controller than a corrected, quantified account of why surrogate-trained controllers succeed or fail, together with a reusable role-separating recipe. The practical implication for building-control engineering is that a surrogate intended as an RL training environment should be validated on downstream closed-loop utility and on the smoothness of its action–response surface, not on predictive error alone. Future work follows from the limitations: multi-zone and mixed-topology buildings; a loss-landscape analysis along the training trajectory; a validated rather than auxiliary heating-power channel; transfer beyond the hydronic family, pairing the transferable calibration pipeline with light policy adaptation instead of zero-shot reuse; and, ultimately, deployment on a physical building.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The numerical artifacts supporting the reported figures and tables are documented in the Supplementary Material through provenance maps; the project codebase is not publicly released at this stage. Additional non-confidential derived artifacts may be made available by the corresponding author upon reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with language editing and L^AT_EX formatting. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No generative AI tool was used to generate, analyze or interpret the research data or to produce the scientific findings.

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